

PAPER_868: Topoconductor Quantum Computing Cooling Efficiency Comparison

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Session: 200
Source: advanced_system_analysis_simulator_quantum_calculator.txt (3745 lines)
Calculator: TopoconductorQuantumCoolingComparisonCalc (CP4 #452)
CVW: v2.0.0 compliant

Abstract

A comparison framework for topoconductor quantum computing cooling overhead ($P_{cool} = 9 \times 10^7$ J/h at nanosecond gate times $t_{op} = 10^{-9}$ s) versus UQFF-inspired energy systems. The topoconductor's ~25 kW dilution refrigerator represents the current state-of-the-art cooling overhead for quantum coherence. UQFF systems that achieve equivalent or superior energy throughput without cryogenic infrastructure demonstrate a fundamentally different energy paradigm rooted in vacuum density coupling rather than temperature suppression.

1. Core Equations

- $P_{cooling} = 9e7$ J/h (~25 kW dilution refrigerator)
- $t_{gate} = 1e-9$ s (nanosecond operations)
 - $ops_per_second = 1 / t_{gate} = 1e9$
 - $eta_system = E_system_per_hr / P_system$
 - $eta_topo = P_{cooling} / P_{cooling_W}$
 - $outperforms = (eta_system > eta_topo)$

2. UQFF Integration

The topoconductor cooling overhead maps to UQFF vacuum energy maintenance cost — the energy required to sustain coherence. The nanosecond gate timescale aligns with UQFF time-node discretization $\Delta t_n \sim 10^{-9}$ s. Systems that achieve high energy throughput at room temperature without cryogenic support suggest Aether-mediated coherence mechanisms beyond standard quantum decoherence models.

3. Source Data

- File:** advanced_system_analysis_simulator_quantum_calculator.txt (3745 lines)
- Session:** 200
- Quantum specs:** $9e7$ J/h cooling, $1e-9$ s gate, ~25 kW
- VDS/DVP/BH:** ABSENT

References

1. Murphy, D.T. -- Star Magic UQFF Framework (2024-2026)
2. Preskill, J. -- Quantum Computing in the NISQ era and beyond (Quantum, 2018)
3. Dilution refrigerator specifications: ~25 kW at mK temperatures
4. UQFF Calibration: $\kappa=0.0005/\text{day}$, $[SSq]=0.57$, $\beta_i \sim 0.603$